Contextual Write–Read–Value Attention: An Architecture-Path Study

Anonymous authors
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Abstract

We study contextual Write–Read–Value (W/R/V) attention for an approximately 98.2M-parameter causal language model with tied lexical residual objects and deterministic micro-experts. W/R/V uses eight contextual read heads, two shared contextual write/value heads, per-head RMS normalization, rotary position encoding, and a zero-initialized lexical residual in each write stream. PyTorch scaled dot-product attention is configured to permit FlashAttention dispatch for the resulting causal softmax. A completed 100M-token run reaches 4.706118 evaluation loss at 118,416 logged training tokens/s on one H100. An archived grouped-query-attention run reaches 4.820476 at 111,355 tokens/s, but it used a lower peak learning rate (3×10^{-4} versus 4×10^{-4}) and lacks a colocated realized configuration; we therefore report the numerical difference without claiming an aligned architectural win. Fixed-state WVR ablations remain informative negative controls: their best completed result is 4.895427. The evidence supports the complete realized W/R/V configuration as a candidate for controlled replication, not an isolated lexical-residual effect.

1 Introduction

Attention-based language models combine contextual transport with content-dependent selection over individual past token representations. We study whether this selection can be parameterized as contextual Write–Read–Value (W/R/V) attention while injecting token identity only through a narrow, tied lexical residual.

Our design follows a shared-plus-residual principle. Every token traverses the same dense computation path. Token identity modulates narrow lexical-object and micro-expert branches, and a zero-initialized lexical write residual is tied to the same rank-16 table. The main sequence mixer is causal W/R/V attention with eight contextual read heads and two shared contextual write/value heads. Per-head RMS normalization and rotary position encoding precede PyTorch scaled dot-product attention.

The architecture emerged from a systematic fixed-state WVR study. Those recurrent variants are consistent with a limitation of a single compressed matrix for instance-selective retrieval. We retain them as negative path ablations rather than presenting them as the final architecture.

Our contributions are empirical:

- contextual W/R/V attention with grouped read/write heads, per-head RMS normalization, RoPE, and SDPA configured to permit FlashAttention;
- an implementation of a tied, zero-initialized lexical residual on contextual write streams, alongside tied lexical feed-forward objects and deterministic micro-experts;
- a completed 100M-token W/R/V result at 4.706118 held-out loss and 118,416 tokens/s;
- fixed-state WVR and attention-path ablations that motivate, but do not isolate, hypotheses about compressed and lexical addressing; and

- raw-artifact-backed reporting that separates completed observations from optimizer-matched architectural claims.

The observed W/R/V run is numerically better and faster than our archived GQA run, but the peak learning rates differ. We therefore do not yet claim an optimizer-matched win. The required confirmation is a GQA rerun at 4×10^{-4} with the same harness, plus saved final checkpoints and direct lexical-residual ablations.

2 Method

2.1 Contextual W/R/V attention

Let $\boldsymbol{H} \in \mathbb{R}^{B\times T\times d}$ be normalized hidden states. Each layer projects contextual reads, writes, and values,

$$\boldsymbol{R} = \boldsymbol{H}\boldsymbol{W}_r, \qquad \boldsymbol{W}_{\text{ctx}} = \boldsymbol{H}\boldsymbol{W}_w, \qquad \boldsymbol{V} = \boldsymbol{H}\boldsymbol{W}_v. \tag{1}$$

The evaluated model uses eight read heads and two shared write/value heads, with head dimension 48. A deterministic token code and a learned projection of the tied rank-16 lexical table form a lexical write address $\boldsymbol{L}(x_t)$. For write head h,

$$\boldsymbol{w}_{t,h} = \boldsymbol{w}_{t,h}^{\text{ctx}} + \tanh(g_h)\,\boldsymbol{L}_h(x_t), \qquad g_h = 0 \text{ at initialization.} \tag{2}$$

Thus the lexical branch is exactly neutral initially; the contextual write stream remains fully active. Reads and writes receive independent RMS normalization per head and rotary position encoding. Write/value heads are repeated across their four read-head groups, and causal attention is

$$\boldsymbol{y}_{t,h} = \sum_{j\le t} \text{softmax}_j\left(\frac{\text{RoPE}(\boldsymbol{r}_{t,h})^\top\,\text{RoPE}(\boldsymbol{w}_{j,\kappa(h)})}{\sqrt{48}}\right)\boldsymbol{v}_{j,\kappa(h)}, \tag{3}$$

where $\kappa(h)$ maps each read head to one of the two shared write/value heads. PyTorch scaled dot-product attention is configured to permit FlashAttention dispatch on H100. W/R/V is therefore an attention mechanism; the terminology describes data flow and the lexical write residual, not the absence of softmax attention. Incremental execution retains past W and V tensors and grows with prefix length.

2.2 Tied lexical residual object

The shared feed-forward transformation is a dense SwiGLU,

$$\boldsymbol{F}(\boldsymbol{h}) = \boldsymbol{W}_d\left[\text{SiLU}(\boldsymbol{W}_g\boldsymbol{h}) \odot \boldsymbol{W}_u\boldsymbol{h}\right]. \tag{4}$$

For token identity x_t, a tied table $\boldsymbol{E} \in \mathbb{R}^{|\mathcal{V}|\times r}$ provides a low-rank modulation:

$$\boldsymbol{z}_t = \text{SiLU}(\boldsymbol{U}\boldsymbol{h}_t) \odot (\boldsymbol{1} + \boldsymbol{E}[x_t]), \tag{5}$$

$$\boldsymbol{O}(\boldsymbol{h}_t, x_t) = \boldsymbol{D}\boldsymbol{z}_t. \tag{6}$$

The same table $\boldsymbol{E}$ is shared across layers; projection matrices remain layer-specific. Setting $\boldsymbol{E}$ to zero makes the token-dependent modulation neutral initially, but the object branch itself remains active through a nonzero output gate α:

$$\boldsymbol{F}_{\text{object}}(\boldsymbol{h}_t, x_t) = \boldsymbol{F}(\boldsymbol{h}_t) + \alpha\boldsymbol{O}(\boldsymbol{h}_t, x_t). \tag{7}$$

2.3 Deterministic micro-expert residual

Each layer assigns token x_t to one of N micro-experts using

$$e_\ell(x_t) = (x_t + \ell) \bmod N. \tag{8}$$

All expert hidden activations are computed as one regular tensor operation and the selected output is gathered without sorting or scatter. With expert width w,

$$\boldsymbol{R}_\ell(\boldsymbol{h}_t, x_t) = \beta\, \boldsymbol{W}_d^{(e_\ell(x_t))} \left[\mathrm{SiLU}(\boldsymbol{W}_g^{(e_\ell(x_t))}\boldsymbol{h}_t) \odot \boldsymbol{W}_u^{(e_\ell(x_t))}\boldsymbol{h}_t\right]. \tag{9}$$

The complete feed-forward output is $\boldsymbol{F} + \alpha\boldsymbol{O} + \beta\boldsymbol{R}$. This is a lexical residual over a shared dense path, not a claim that routing replaces shared computation.

2.4 Fixed-state WVR associative ablation

The earlier fixed-state ablation uses one recurrent associative matrix rather than softmax attention. We denote that mechanism by *WVR*: write address, contextual value, and read address. For token x_t and hidden state $\boldsymbol{h}_t$, a compressed lexical forge reuses the tied lexical object table,

$$\boldsymbol{f}_t = \boldsymbol{W}_{f,2}\,\mathrm{SiLU}(\boldsymbol{W}_{f,1}\boldsymbol{E}[x_t]), \tag{10}$$

where the implemented bottleneck is $16 \to 4 \to 128$. The common address combines this deterministic lexical code with a projected hidden-state signature,

$$\boldsymbol{c}_t = \mathrm{norm}(\boldsymbol{W}_c\boldsymbol{h}_t), \tag{11}$$
$$\boldsymbol{a}_t = \mathrm{norm}(\mathrm{forge}(x_t) + \tanh(\eta)\boldsymbol{c}_t)\,. \tag{12}$$

The lexical code is an address tied to token identity; it is not evidence of semantic representation. The contextual projection is likewise described operationally rather than semantically.

Writing uses the preceding address and the current contextual payload,

$$\mathbf{w}_t = \boldsymbol{a}_{t-1}, \qquad \mathbf{v}_t = \boldsymbol{W}_v\boldsymbol{h}_t, \qquad \mathbf{r}_t = \boldsymbol{a}_t, \tag{13}$$
$$\widehat{\mathbf{v}}_t = \mathbf{w}_t^\top\boldsymbol{M}_{t-1}, \tag{14}$$
$$\boldsymbol{M}_t = \lambda\boldsymbol{M}_{t-1} + \rho\mathbf{w}_t(\mathbf{v}_t - \widehat{\mathbf{v}}_t)^\top, \tag{15}$$
$$\boldsymbol{y}_t = \mathbf{r}_t^\top\boldsymbol{M}_{t-1}, \tag{16}$$
$$\boldsymbol{c}_t^{\mathrm{WVR}} = \gamma\boldsymbol{W}_o\boldsymbol{y}_t. \tag{17}$$

The context residual $\boldsymbol{c}_t^{\mathrm{WVR}}$ is added to the causal-convolution output. The implementation shifts the write stream causally: the current read uses $\mathbf{r}_t$, while the preceding write address is paired with the current payload. The chunkwise triangular solve is an exact parallelization of this recurrence. Thus WVR stores a contextual hidden value rather than a lexical code. The state is one rank-128 matrix plus fixed-size address and collision-load vectors, independent of prefix length.

Collision normalization maintains a decayed diagonal load estimate $\boldsymbol{d}_t$ and scales write and read coordinates by $(\boldsymbol{d}_t + \epsilon)^{-1/2}$ before normalization. It does not partition the rank-128 matrix into separate memories.

QKV attention compares a contextual query with every retained key and performs normalized competition over individual token occurrences. WVR instead reads the pre-update compressed recurrent association through $\mathbf{r}_t^\top\boldsymbol{M}_{t-1}$. Our training comparison concerns modeling quality and throughput without assigning credit for KV-cache size. The representational tradeoff is explicit: WVR has fixed state, but repeated or colliding occurrences may overwrite one another.

The experimental occurrence-address extension replaces the dense contextual projection by a compact $d \to 8 \to 128$ signature and adds a gated deterministic position signature,

$$\boldsymbol{a}_t = \mathrm{norm}(\mathrm{forge}(x_t) + \mathrm{smallctx}(\boldsymbol{h}_t) + \tanh(\gamma)\,\mathrm{pos}(t))\,, \tag{18}$$

with $\gamma = 0$ initially and one fixed-size position counter for incremental decoding. Its completed full-budget run is reported as a negative ablation.

2.5 Architectural path ablations

We retain three completed intermediate conditions: one W/R/V layer inserted into a collision-normalized WVR model; eight W/R/V layers with strongly lexical normalized addresses; and eight contextual W/R/V layers with two remaining WVR layers. These are path ablations, not isolated factorial controls. Their ordering motivates hypotheses about insufficient selective attention, overly dominant lexical addressing, and residual fixed-state bottlenecks, but does not identify those causes independently.

3 Experimental Design

3.1 Controls and architectural path

The study compares GQA, fixed-state associative variants, hybrid WVR–attention models, and the final ten-layer contextual W/R/V model at approximately 98.2M parameters. Fixed-state conditions include the additive predecessor, Stable Delta, collision normalization, multiple timescales, lexical values, Lexical Forge V3, Forge V3 plus collision normalization, and compact context-plus-position addressing. The archived attention path evaluates one W/R/V layer, eight strongly lexical W/R/V layers, eight contextual W/R/V layers with two WVR layers, and the final ten-layer contextual W/R/V model with per-head RMS normalization.

The historical causal-convolution lexical micro-expert run trained for 1,000,013,824 tokens is reported separately. It is not a matched control for the 100M-token WVR table.

3.2 Data, tokenizer, and budget

Models use a 200,019-entry tokenizer recorded as `o200k_base` and FineWeb-Edu sample-10BT. The tokenizer asset checksum was not archived and remains a reproducibility blocker. The principal runs use sequence length 2,048, seed 42, BF16, and 100,007,936 training tokens. A deterministic 5% document stream is reserved for evaluation and excluded from training by the data pipeline. Each final estimate averages 8 batches, or 262,144 tokens; no uncertainty estimate is available. Table 1 reports the last finite evaluation, normally step 3,000.

The archived GQA metrics used peak learning rate 3×10^{-4}; recent WVR and contextual W/R/V runs use 4×10^{-4}. No realized GQA configuration is colocated with its CSV, so dimensions, exact token budget, tokenizer, dataset construction, seed, software path, and kernel settings cannot be independently reverified from the review artifact. Cross-run comparisons are descriptive rather than controlled architecture comparisons.

3.3 Metrics and evidence rule

We report held-out next-token cross-entropy, realized parameter count, exact token count, and the training throughput logged at the finite evaluation step. Quality and throughput are separate claims. Every numerical table entry must map to an archived CSV or JSON artifact; generated figures are derived from those files and are not primary evidence. Short proxies are used only as failure screens because several favorable proxy rankings did not transfer to full budget.

3.4 Context diagnostics

Associative recall, induction, and synthetic in-context probes are evaluated by distance. They test instance-selective retrieval that average language-model loss may hide. The completed attention-free checkpoints score zero on the archived recall and induction probes at all tested distances. The available GQA synthetic artifact is not sufficiently documented as the same protocol, so no numerical positive-control claim is made. Persistent-state claims beyond the convolutional receptive field remain unsupported and require dedicated 4–8k-context tests.

Table 1: Completed nominal 100M-token results read from archived CSV/JSON measurements. Throughput is one logged point at the final finite evaluation step on one H100. The GQA row lacks a realized configuration and used peak LR 3×10^{-4}; recent WVR and W/R/V runs used 4×10^{-4}.

Mechanism	Held-out loss ↓	Train tok/s ↑
Dense GQA (QKV)	**4.820476**	111,355
Shared additive associative context	4.921405	124,363
Stable Delta WVR	4.898880	121,971
Multi-timescale Delta WVR	4.905554	120,496
Collision-normalized WVR	4.895427	122,282
Collision WVR + 1 lexical W/R/V layer	4.892026	123,145
Lexical-value WVR	4.963167	123,113
Lexical Forge V3 WVR	4.900726	121,697
Forge V3 + collision normalization	4.897241	121,312
Compact context + position address	4.899072	122,764
Strongly lexical W/R/V, 8/10 layers	4.973798	118,546
Contextual W/R/V, 8/10 layers	4.852761	118,936
Contextual W/R/V + RMSNorm, 10/10	**4.706118**	118,416

3.5 Hardware and numerical checks

The full runs use one NVIDIA H100 80GB HBM3. FP32 unit tests cover causality and full/incremental equivalence for contextual W/R/V, plus fixed state and stable state addresses for convolutional and recurrent WVR primitives. Archived BF16 checkpoint profiles are reported separately and do not support an exact 10^{-5} checkpoint-level equivalence claim. The final W/R/V run disabled periodic checkpoint saving; its metric and realized configuration artifacts are archived, but learned gate values cannot be recovered. Generic top-k/Zipf fields in the serialized CLI namespace are inactive for the realized `lexical_object_micro_expert` module, whose actual assignment is the modulo rule in Section 2.

4 Results

4.1 Completed 100M-token architecture path

Collision-normalized WVR is the best completed fixed-state associative variant, improving Stable Delta by 0.003453 loss. Adding one W/R/V attention layer changes loss by 0.003401. The strongly lexical eight-layer condition reaches 4.973798, while the contextual eight-layer hybrid reaches 4.852761. The final jointly changed condition removes the two remaining WVR layers and adds per-head RMS normalization, reaching 4.706118 at 118,416 tokens/s. These path comparisons do not isolate lexical residuals, normalization, or layer replacement as causes.

The final row is 0.114358 lower in loss, and its single logged throughput point is 6.34% higher than the archived GQA point. This is not an architectural superiority or reproducible speedup estimate: peak learning rate differs and the GQA realized configuration is missing. A fully archived GQA rerun at 4×10^{-4} is required before declaring a controlled winner.

The lexical forge reuses the rank-16 lexical-object table through a $16 \to 4 \to 128$ network. Its strong short-proxy improvement did not transfer: Forge V3 reached 4.900726 at full budget, and combining it with collision normalization reached 4.897241 rather than improving the collision-only result. A lexical payload was substantially worse at 4.963167, supporting the design decision to retain $\mathbf{v}_t = \boldsymbol{W}_v \boldsymbol{h}_t$. Independent and weakly coupled read/write forge variants were rejected in paired proxies; their raw JSON artifacts are retained as negative ablations rather than promoted to the canonical implementation.

4.2 Proxy reliability

Short matched proxies were useful for detecting catastrophic failures but were not reliable rankers of final 100M-token loss. Multi-timescale Delta, Lexical Forge V3, and Forge V3 plus collision normalization each looked favorable in a short proxy and failed to improve the corresponding full-budget control. Consequently, no architectural claim in Table 1 is based on a proxy. The compact context-plus-position occurrence address also failed to improve collision normalization at full budget (4.899072 versus 4.895427).

4.3 Longer-budget convolutional run

The separate historical 1B-token attention-free convolutional run contains 98,194,244 parameters and reached held-out loss 4.180625 at step 30,000 with measured throughput 113,621 tokens/s. Because its token budget differs by a factor of ten, it is not used to rank the 100M-token QKV and WVR mechanisms.

4.4 Incremental equivalence and fixed state

FP32 unit tests reproduce full-sequence outputs under token-by-token incremental execution at tolerance 10^{-5} for the convolution and WVR primitives. The archived BF16 checkpoint profiles show larger numerical differences (approximately 0.02–0.17 maximum absolute error), so exact checkpoint-level BF16 equivalence is not claimed. For WVR, the recurrent state is one rank-128 association matrix, the previous write address, and—for collision normalization—one rank-128 diagonal load vector. The occurrence-address ablation adds one scalar absolute-position counter. These state shapes do not grow with generated context length.

We audit the realized exported convolutional checkpoint rather than inferring the absence of attention from configuration alone. Across 182 tensors, the name-and-shape audit finds no query, key, value, or fused-QKV projection. Compatibility paths such as `self_attn` do not correspond to an attention operation. This structural audit is not treated as a substitute for runtime tests.

The archived associative-recall and induction probes report zero accuracy for the evaluated attention-free checkpoints at every tested distance. The available GQA synthetic artifact does not document an identical protocol, so we do not report a quantitative positive-control comparison. The negative attention-free diagnostic alone does not establish instance-selective retrieval.

5 Related Work

Transformers use contextual query–key competition and value aggregation (Vaswani et al., 2017). Grouped-query attention shares key/value heads across query groups to reduce memory and bandwidth (Ainslie et al., 2023). The central W/R/V operation is in this family: read heads play the query role, shared write heads play the key role, and values remain contextual. We therefore do not claim that renaming the streams creates a new attention operator. The architectural distinction studied here is a tied, zero-initialized lexical residual on contextual writes, integrated with lexical feed-forward residuals.

The final model also uses established components: rotary position encoding (Su et al., 2021), per-head read/write normalization closely related to query–key normalization (Henry et al., 2020), and exact IO-aware attention kernels (Dao et al., 2022). Our empirical question is whether their combination with tied lexical residuals changes the quality–throughput tradeoff under a small-model budget.

RWKV, RetNet, and Mamba develop recurrent or state-space alternatives with linear-time sequence processing (Peng et al., 2023; Sun et al., 2023; Gu & Dao, 2023). Our fixed-state WVR ablations are closer in motivation than the final W/R/V model, but they use a distinct single associative matrix and perform worse in the present controls. The final model abandons fixed-state decoding and retains grouped W/V caches; it should not be presented as a linear-state competitor to those methods.

6 Limitations and Broader Impact

The current study is limited to approximately 100M parameters, one tokenizer, one web-text corpus, and one seed for the principal runs. A favorable result does not establish scaling behavior or universal superiority to GQA. Hardware throughput is implementation- and device-specific and must not be generalized from one H100.

The current study does not establish production inference throughput. W/R/V retains W/V tensors whose size grows with context, like a grouped KV cache. Long-context prefill, time to first token, serving latency distributions, and CUDA Graph replay remain outside the present claims.

Token-identity conditioning may encode corpus-specific frequency and lexical biases. FineWeb-Edu inherits limitations of large web corpora, including uncertain provenance, representational imbalance, and potentially harmful content. The proposed architecture does not remove these risks. Efficiency gains can also lower the cost of deploying low-quality or harmful models. We report data provenance, held-out construction, and deployment constraints so these tradeoffs can be assessed.

Fixed-state WVR does not retain individually addressable past token occurrences. Repeated lexical addresses and projected-context collisions are compressed into one recurrent matrix. Collision normalization mitigates coordinate saturation but does not recover overwritten instances. These negative controls motivate contextual W/R/V attention; they are not properties of the final model.

The strongest comparison is not optimizer-matched: GQA used peak LR 3×10^{-4} and final W/R/V used 4×10^{-4}. The final run also saved no checkpoint, preventing post-training inspection of lexical write gates. Consequently, the current evidence cannot isolate gains from learning rate, per-head normalization, implementation details, or the lexical residual. Required controls are GQA at 4×10^{-4} and W/R/V with the lexical residual fixed off, both with saved checkpoints and preferably multiple seeds. Finally, short proxies were not reliable predictors of full-budget ordering; they should remain failure screens rather than winner selectors.

7 Conclusion

We evaluated a complete contextual W/R/V configuration with grouped read/write heads, per-head RMS normalization, RoPE, SDPA, tied lexical residual objects, and deterministic micro-experts. The completed 98,195,224-parameter run reaches 4.706118 evaluation loss at a logged point of 118,416 training tokens/s. This is numerically lower than the archived GQA result of 4.820476 at 111,355 tokens/s, while fixed-state collision-normalized WVR reaches 4.895427.

The numerical ordering is encouraging but not a controlled superiority claim: GQA used a different learning rate and lacks a realized archived configuration. No final W/R/V checkpoint was saved, so learned lexical gates cannot be audited. The immediate requirements are a fully archived GQA rerun, a W/R/V lexical-write-off control, factorial layer/normalization controls, saved checkpoints, and repeated throughput measurements. The fixed-state experiments remain useful negative path evidence.

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A Reproducibility Checklist

The repository snapshot accompanying this revision contains:

- model source and realized JSON configurations for the recent WVR and contextual W/R/V conditions; the GQA row has metrics but no realized archived configuration;
- tokenizer name, vocabulary size, dataset identifier, deterministic evaluation ratio, seeds, sequence lengths, learning rates, and exact token budgets;
- archived per-step CSV metrics or JSON measurements for every table row, including negative ablations;
- deterministic collection and plotting scripts for the checkpoint and context diagnostics;
- unit tests for contextual W/R/V causality, grouped head shapes, tied lexical-table identity, neutral lexical initialization, RoPE, and FP32 full/incremental equivalence, plus fixed-state WVR tests;
- a name-and-shape checkpoint audit for the historical attention-free control, explicitly not applied to the final attention model.

Large checkpoints, Git history, credentials, cloud addresses, hostnames, local user paths, and transient logs are excluded from the review artifact. No final checkpoint exists for the reported best W/R/V run because checkpoint saving was disabled; this prevents post-training gate inspection. The run also records no Git commit. The reviewed source fixes post-run causal-mask/chunked-cache handling and residual-output initialization; it is an audit-corrected implementation, not a byte-identical historical snapshot, and the reported result requires replication. Raw run configurations are scrubbed only for identifying paths, which are replaced by `[REDACTED]`. The double-blind supplement must be built as a separately scanned archive without `.git` metadata or repository links.